

L02: System Taxonomy and Paradigm Selection

Simulation Without a Black Box

Outline

Learning Objectives

State Space: Discrete vs. Continuous

Four Simulation Paradigms

Worked Examples

Cost of Complexity

Activity

Key Takeaways

Learning Objectives

By the end of this lecture you will be able to:

- Distinguish discrete vs. continuous state spaces and explain which DES assumes.
- Compare four major simulation paradigms: DES, System Dynamics, ABM, and Monte Carlo.
- Map realistic systems (ED, water network, traffic, portfolio) to the appropriate paradigm.
- Apply the “cost of complexity” principle to avoid over-engineered models.

State Space: What Changes Between Events?

Discrete state

State takes on countable values; changes instantaneously at event times.

- Queue length: $0, 1, 2, \dots$
- Inventory level (units in stock)
- Number of busy servers

Continuous state

State varies smoothly over time, governed by differential equations.

- Tank fluid volume $V(t)$
- Blood glucose level $g(t)$
- Room temperature $T(t)$

Key implication for DES

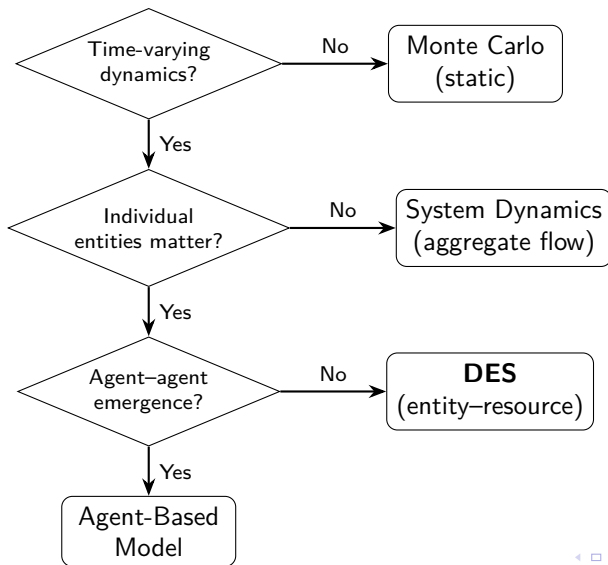
Between events the system state is **constant**. This is what makes event-driven time advance efficient — we jump over idle periods.

Four Paradigms at a Glance

Paradigm	State	Time advance	Interactions	Best for
DES	Discrete	Event-driven	Entity–resource	Operations, logistics
System Dynamics	Continuous	Differential	Stock–flow	Policy, macro trends
Agent-Based (ABM)	Mixed	Fixed step	Agent–agent	Emergence, behavior
Monte Carlo	N/A	Static	Independent draws	Risk, integration

No paradigm dominates all problems. The choice is driven by the *question*, not the modeler's preference.

Paradigm Selection Flowchart



Worked Example 1: Hospital Emergency Department

System: Patients arrive, triage, see physician, receive treatment, discharge.

Key questions:

- 90th-percentile door-to-physician time?
- How many beds are required at 90% occupancy?

Why DES?

- Patients are discrete entities with attributes (acuity, arrival time).
- Physicians and beds are capacitated resources.
- Time-varying arrival rates (peaks at 10 am and 6 pm).
- Question is about operational flow under

Rejected alternatives

SD: cannot track individual patient paths.

MC: ignores sequencing and congestion.

M/M/c formula: assumes Poisson arrivals + exponential service; no acuity classes, no time-varying rates.

Worked Examples 2–4

Water distribution network

Flows governed by PDEs; pressure depends on pipe geometry and topology.

Paradigm: ODE/PDE solver.

Continuous state; no discrete events; deterministic.

Urban traffic (macro)

Aggregate vehicle density evolves over the network.

Paradigm: System Dynamics.

Individual cars don't matter; flow equations dominate. (Micro-traffic → ABM.)

Investment portfolio

One-year horizon; independent daily return draws; terminal wealth.

Paradigm: Monte Carlo. Static (no queue); exact independence; no congestion effects.

The Cost of Complexity Principle

Principle (Law of Parsimony)

A model should be exactly as complex as the decision requires — no simpler, no more complex.

Under-scoped model

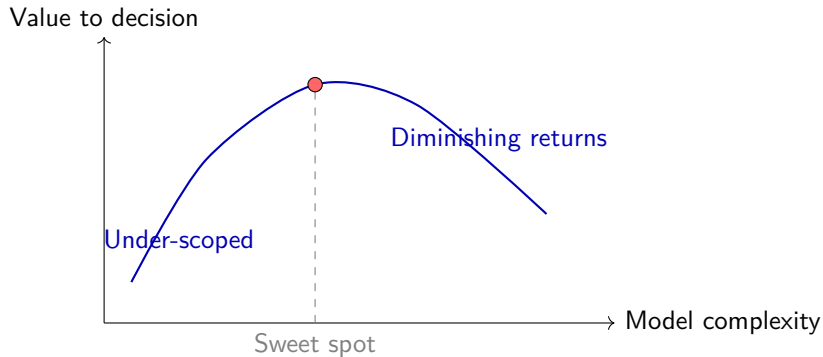
- Ignores key interactions.
- Produces confident but wrong answers.
- Example: M/M/1 formula for a multi-class, time-varying ED.

Over-scoped model

- Requires data that do not exist.
- Long build and debug time.
- Hard to validate; stakeholders distrust it.
- Example: full ABM when a single-queue DES suffices.

*The model is a **tool for a decision**, not a replica of the world.*

Complexity vs. Decision Value



Aim for the inflection point: sufficient fidelity to answer the decision, not more.

Activity: Map Five Real Systems to Paradigms

For each system: (a) identify the dominant characteristics, (b) choose a paradigm, (c) justify in one sentence.

1. University registrar's course-enrollment system on the first day of add/drop.
2. Spread of an infectious disease through a city of 500,000.
3. A manufacturing line with 8 stations and random machine breakdowns.
4. Option pricing under the Black–Scholes model.
5. National power grid with renewable generation and stochastic demand.

Be prepared to defend your choice. There may be more than one defensible answer.

Key Takeaways

- Discrete + stochastic + dynamic = the DES niche.
- System Dynamics, ABM, and Monte Carlo each have their own niche — know all four.
- The paradigm choice is driven by the *question*, not habit or familiarity.
- Complexity has diminishing returns: more detail costs build time and validation effort.
- DES is most powerful when individual entity paths and resource contention both matter.

Next lecture: Conceptual modeling — defining system boundaries (L03).